

When Artificial Intelligence meets Social Sciences and Humanities groups in Brazil, 2022

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Abstract

Our research focuses on the emergence of the Brazilian research ecosystem regarding humanities and social sciences in relation to Artificial Intelligence (AI) technologies. Using computational web scraping and topic modeling techniques, we identified the most frequent topics studied by the research groups registered in the CNPq’s Directory of Research Groups, depending on the scientific area and type of technology of interest. Business Administration and Law stand out as the main areas which research the social impacts of AI, while Linguistics has produced Natural Language Processing techniques. Another result is the frequent use of the term Artificial Intelligence to describe uses of non-intelligent digital technologies.

Keywords: Artificial Intelligence, Humanities; Social Sciences, scientometrics, Brazil

1 Introduction

Our research focuses on the emergence of the Brazilian research ecosystem regarding humanities and social sciences in relation to Artificial Intelligence (AI) technologies, with the objective of understanding which topics are attracting the most attention from the Brazilian scientific community and what this could tell us about the human resources’s training in AI in the country.

Artificial Intelligence (AI) corresponds to a class of high-processing technologies that — through machine learning — refine predictive and automated algorithms capable of solving complex problems (Boden, 1996). Because of its capabilities, AI has had an impact on different social domains, such as education (Selwyn, 2022; Ouyang & Jiao, 2021); the economy and labor market (Mucha & Seppala, 2020; Noman, Akter, Pranto, & Haque, 2022); politics and social media (Ajala et al., 2022; Hudson, 2019); law and public safety (Cárdenas & Molano, 2021), culture (Airolidi, 2021), among others (Esposito, 2022).

This is progressively increasing the scientific community’s interest in explaining or understanding what these AI technologies are, therefore, leading universities, educational

and research institutions and specialized centers (such as *think-tanks*) to mobilize scientists and engineers to study them. It's no different with humanities experts and social scientists.

2 Methods

2.1 Data collection procedures

The starting point of the research was an initial list of descriptors provided by another team from the Center for Artificial Intelligence (C4AI) at the University of São Paulo. The initial list of descriptors (Appendix A) was complemented by a bibliographic survey of Artificial Intelligence technologies and their relationship with humanities and social sciences. It is important to note that this step also aimed to find new terms that could have been missing from the list initially provided.

In December 2022 ¹, each descriptor was used in the parameterized search in order to find a match with the name of the group (or) the name of the line of research (or) the keywords of the line of research. All group names were recorded and duplicates were eliminated. In all, 1547 research groups were found.

The next step was to use computational techniques to scrape the data from each group, using the Selenium, Requests, BeautifulSoup, Pandas and Pickle libraries in the Python programming language. The first one is responsible for automating computer commands, with the aim of simulating human actions, i.e. it should enter the DGP, search for and access each of the 1547 groups. While Requests allowed the manipulation of each group's data, the BeautifulSoup and Pandas libraries scraped and organized each group's information into dataframes. Finally, Pickle stored non-string variables in this same database. The database was then processed and made consistent, mitigating scraping errors.

For each group, we gathered the information about: (1) its institution, (2) its administrative category, (3) the region and federative unit, (4) the group's title, (5) its main and secondary research leaders, (6) the group's area and sub-area, (7) the main leader's area, (8) the impact text, i.e. the group's own description of its activities. We also scraped the research lines ($n = 7741$), researchers ($n = 18181$) and students ($n = 31999$) associated with each group.

Next, we filtered out and selected only the 'Human Sciences', 'Applied Social Sciences' and 'Linguistics, Languages and Arts' groups - categories given by the CNPq itself. We also applied a second filter to exclude groups that had been registered in the scraping, but did not have any type of research in new digital technologies and Artificial Intelligence. The number of groups was reduced to 196. Consequently, so were the research lines ($n = 775$), researchers ($n = 1836$) and students ($n = 2100$). This number involves all the researchers and students who participate or have participated in the groups.

Finally, pairs of researchers were responsible for classifying whether each group, according to their line of research, impact text or title, was:

- *Researching Artificial Intelligence as an object.* The group researches the social, ethical, philosophical, moral, historical, linguistic, artistic, cultural, political and economic causes and effects of AI technologies ($n = 55$).
- *Using Artificial Intelligence as a technique.* The group uses computational techniques and AI tools to study other phenomena, or seeks to produce such tools ($n =$

¹The date is relevant because the data collection was carried out at the same time as the ChatGPT release.

42).

- *Both researching Artificial Intelligence as an object and using it as a technique.* In this case, the group both produces or uses AI computational techniques and studies their implications for a given research object ($n = 33$).
- *Concerning exclusively Digital technologies.* The group does not focus on Artificial Intelligence - either as a technique or as an object - but is interested in other digital technologies ($n = 66$).

The impact texts and the research lines of each group were lemmatized, which consists of representing words using the infinitive of verbs and the masculine singular of nouns and adjectives (De Lucca & Nunes, 2002). This tends to remove later errors in the analysis and systematization of the results. Both ChatGPT-3.5 and Spacy (a Python library) helped with this task.

2.2 Analysis and systematization procedures

Topic modeling is a Natural Language Processing technique, with the advantage to allow to get an overview of the contents that are being discussed in our corpus. NLP names a whole class of generative probabilistic models, as Latent Dirichlet Allocation (LDA). In this research, we use Correlation Explanation (CorEx), which operates with semi-supervised machine learning, in groups's research lines (Hovy, 2020).

We then transformed the words into vectors so that the CorEx library could group them using "anchor" words, i.e. we provided guide words so that the AI could assemble the topics according to those closest to our references. In this way, we came up with five topics based on a combination of the words "development and innovation", "economy and work", "government and public policies", "teaching and education" and "institution" — the term we associate with private capital companies. These anchors were chosen by counting the words with the most appearances in the research lines and the groupings were made with words with a frequency greater than or equal to two.

As Figure 2 shows, the first topic tells us that the development area is guided by sustainability, as well as issues related to industry, entrepreneurship and health. Ethical issues are also present in Topic 5, which points to private capital's concern with issues relating to culture and agriculture. Topic 2 deals with work and the economy, showing that these themes are related to issues of surveillance, governance and private capital, whether from a global perspective ("development" and "globalization") or a local one ("startup"). Topic 3 points out how economics is also an area of interest for government management, as well as transparency, control and regulation. Finally, Topic 4 addresses the area of education, highlighting the use of artificial intelligence for teaching mathematics.

Topic modeling is complemented by Iramuteq, a textual analysis software. With it, we use similarity analysis and factor analysis. The first, based on graph theory, aims to identify words that appear at the same time with some frequency and then structures them in a graphical network display. The other associates words according to variables of interest, such as administrative category.

3 Results

In order to analyze, systematize and interpret the results below, we must discuss how Artificial Intelligence is researched by each research group, according to its scientific field. There is a classification problem inherent in this exercise, due to the fact that AI has been

Tópico 1. Inovação e desenvolvimento sustentável, empreendedorismo, propriedade intelectual, saúde, indústria, ética, comunicação
Tópico 2. Economia e trabalho instituição, globalização, desenvolvimento, vigilância, governança, decolonial, startup
Tópico 3. Governo e Política Pública economia, transparência, dados abertos, regulação, controle, gestão
Tópico 4. Educação e ensino educação, ensino, matemática
Tópico 5. Instituição socioambiental, agronegócio, política, ética, conflito, transculturalidade, mercado

Figure 2: Topic modeling of wall groups's research lines

predict states of well-being and mental health, whose metrics are provided by big data (). The psychological consequences of violence and virtual predation – beyond interactions between bots and human beings – are also problematized ().

Education ($n = 17$) was one of the few areas where there were groups in all classifications. There is a concern with the school curriculum, investigating the impact of these Artificial Intelligence technologies on learning in STEM courses – especially in Mathematics (). New active learning methodologies are being developed using these technologies to improve teaching, such as educational digital games and simulations (). In addition, there are groups developing tools for mining educational data and allowing it to be visualized (). Finally, digital cultures – in school contexts – are another focus of these groups ().

The Philosophy groups ($n = 5$) research the ethical aspects behind Artificial Intelligence, as well as the ontological and epistemological aspects of these technologies (). The interface with the biological sciences (neuroethics, cognitive sciences and bioethics) is notable (). They also investigate environmental philosophy and information architectures ().

Brazilian geographers ($n = 3$) are concerned with the regional development of the Artificial Intelligence industry (), renewable and sustainable energies (), biotechnology (), the geopolitics of innovation () and the political economy of the city (). The emphasis is on local productive arrangements for these technologies (). In the case of Artificial Intelligence as a method and technique, the groups emphasize geoprocessing techniques and intelligent modeling of environmental and urban systems ().

Archaeologists ($n = 2$) and historians ($n = 12$) do not make the use of AI technologies explicit. They seem to restrict themselves to digital technologies, such as the digitization of documents (), collections and social memories contained in digital environments – with scraping and data mining – (), subjectivities in the digital age (), the history of socio-technical materialities (), interactive archaeology (), and the media in new digital technologies ().

The Arts groups ($n = 4$) are interested in generative Artificial Intelligence technologies, such as the DALL-E platform, which feeds on images available on the internet to generate new combinations between them based on user input (). There are interlocutions with Philosophy when discussing the poetics and aesthetics of new technologies () and contemporary artistic production ().

The Communication groups ($n = 12$) address the issue of digital democracy and pol-

itics, emphasizing how new technologies influence relationships on digital social networks (such as fake news) (). Virtual reality (as in the case of META) (), regulation of the communications sector (), journalism (), digital games (), digital culture and media literacy are also research lines (). There were four Industrial Design groups identified in our scraping. The groups maintain relations with Communication groups, researching how AI methods and techniques can be used to produce accessible interfaces for people with some kind of disability (), for inclusive education () and technologies to support athletes ().

The scraped Modern Languages groups ($n = 6$) do not focus on AI and are in the same situation as the History and Archaeology groups. They use digital technologies for digital philology and textual knowledge representation (). Digital collections and their scraping are also mentioned (). Computer processing of literary works is another type of research ().

This is different among Linguistics groups ($n = 11$). The groups often mention the production of Natural Language Processing (NLP) technologies, which operate with AI (). In addition, the groups research the production of linguistic corpuses to feed NLP models (), the way these technologies alter translation studies and discursive practices (), foreign language teaching (), digital literacy (), biolinguistics and neuroscience () and, ultimately, the philosophy of language ().

Finally, the Architecture and Urbanism groups ($n = 7$) use Artificial Intelligence predominantly as a method. The use for digital visualization is remarkable, especially for historical heritage () and sustainability (). The other uses are restricted to the use of digital technologies and Artificial Intelligence in building and surface infrastructure calculations.

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Now, we will look at the three main areas, Business Administration ($n = 26$), Law ($n = 49$) and Information Science ($n = 19$), which comprise 46.9% of the groups in the sample.

3.1.1 Information Science groups

In Information Science groups, the main debate is about new ways of organizing knowledge using new Artificial Intelligence technologies, i.e. how to preserve digital information and organize it using big data tools. In addition, new methodologies are being developed to deal with increasingly complex webs (see Figure 3).

We found two clusters. To interpret them, a first step is to discuss the associated words within each of the clusters. On one hand, we observe library, knowledge, organization, humanities, representations, language and curatorship. On the other hand, we see complexity, social, network, management, recovery, open, technology and semantics. Then, we suggest to identify the social problems behind this words:

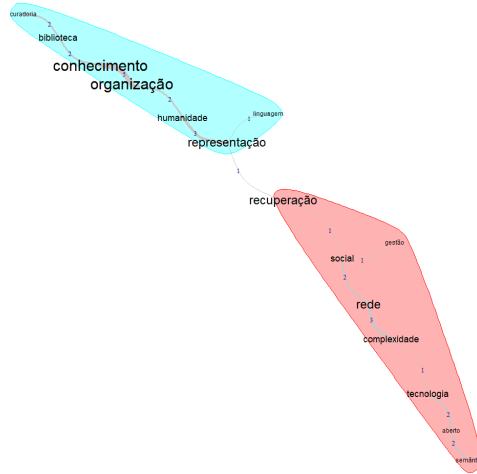


Figure 3: Similitude analysis of Information Science groups

- *Knowledge*: How could big data be used to preserve knowledge in digital libraries situation? How to visualize all this huge volume of data? How to visualize all this huge volume of data?
- *Complexity*: How do information circulate on social networks? How to visualize them?

3.1.2 Law groups

Law has the largest number of semantic clusters (see Figure 1). At first glance, it is striking that the clusters on freedom of speech, the protection of personal data and the circulation of information on the internet seem distant from the others. One working hypothesis is that this debate, restricted to digital technologies, has had little dialog with debates on Artificial Intelligence technologies themselves. Using the similarity analysis technique, we identified clusters on (1) governance, (2) sustainable development, (3) justice, (4) economics, (5) legal democracy.

The first, on governance, is concerned with the production of public policies, the fight against corruption, transparency and the regulation of digital technologies and Artificial Intelligence. The justice cluster is concerned with human rights, digital consumers and access to digital goods and services – and attempts to resolve judicialized conflicts. The sustainable development cluster is related to the previous clusters, but emphasizes how Artificial Intelligence can be a tool for sustainable innovation and micro-entrepreneurship on a global stage. The fourth cluster, on the economy, focuses on the Artificial Intelligence market, taking the market for personal data, personal security and the economic consequences of these technologies as issues. In this cluster on markets, the technical dimension is relevant, as the groups develop Artificial Intelligence technologies to operate in markets. Finally, the last cluster is about legal democracy, emphasizing how the process is carried out.

Then, we suggest systematizing these terms into a social problem that gives unity to the cluster:

- *Governance*: How to mobilize smart technologies to produce efficient public policies?
- *Justice*: How to produce legal instruments to deal with new conflicts that take smart technologies into account?

- *Sustainable development*: How can smart innovations be used to solve environmental problems and mitigate lawsuits?
- *Economy*: How does Artificial Intelligence affect the operation of markets and how can we find ways to regulate them?
- *Legal democracy*: How can Artificial Intelligence technologies be used to speed up legal proceedings in public offices - such as the State?

A closer look shows that the groups researching economics have little association with the groups researching sustainable development. Governance and legal democracy appear at opposite ends of the spectrum, because while one is concerned with streamlining operations within the state bureaucracy, the other focuses on corporate problems. Another finding of the research is that law groups in public institutions emphasize the discussion of justice (and its variations, such as digital justice, digital literacy, human rights, digital democracy) and economic development in the public sector, such as the optimization of bureaucratic processes. Law groups at private institutions, on the other hand, emphasize technological innovation, business development and legal relationships arising from digital environments.

It is important not only to look at the most mentioned terms, but also at those that appeared only once, such as "disability" (whose group is concerned with how Artificial Intelligence can protect basic human rights), "inequality" (and how Artificial Intelligence affects social justice in Latin America), "discrimination" (and how intelligent algorithms may or may not discriminate against people on the internet), "fintechs (and) startups" (to think about how law can offer legislative innovations for these new organizational forms) and "fake news" (around the debate on freedom of expression and legal mechanisms).

3.1.3 Business Administration groups

Whereas Law groups tend to think about the social and legal implications of AI, Business Administration groups turn to business methodology to increase firms' efficiency (see Figure 1). "Inclusion", "biotechnology" and "bioengineering" are some examples of the terms that appear the least (the hapax) in the groups.

The Business Administration clusters are more homogeneous than the Law groups. The orange cluster shows us how smart systems could be used beyond firms, such as in smart cities or agribusinesses. It focuses on transparency and communication issues. The pink cluster brings together expressions that share the problem of what are the implications of Artificial Intelligence for the transformation of contemporary business models. The light blue cluster brings together terms such as supply chain and operations. It is significant that this cluster is the result of the main cluster around the word management. We can infer, with a closer look, that its problematic concerns how Artificial Intelligence affects the logistics operation of supply chains or how it can, on the one hand, make it more efficient and, on the other, how the algorithm can make the best decision in allocating resources. When we examined the main cluster, with the term management, we found that health and Artificial Intelligence has been a topic frequently discussed by Business Administration groups. This cluster points to the implications of AI technological innovations for the management of certain industries, such as the public sector, health and public policy.

4 Discussion

This research on Brazilian AI defines that this scientific field, in the area of humanities, is divided into theoretical and methodological studies, with noticeable variation in scope

according to the knowledge area of each research group. Law stands out in theoretical studies, mainly considering the issue of regulation and ethical impacts, as do philosophers, in AI.

Methodologically, administration stands out, using AI as a means of management and organization for businesses promoted by systems or models, and transformation is also a concern in economics. We also cannot forget about themes that are less mentioned, such as fake news, inclusion, and complexity, which raise alerts for possible research trends. Therefore, even if it occurs in a minority way, humanities areas are present in the scientific production of AI in Brazil, being an incipient area with strong theoretical and methodological insertion.

As our topic modeling and textual analysis guide us, the groups are concerned with the innovation ecosystem (Andion; Alperstedt; Graeff, 2020) and in Brazilian development, which promotes a new research hypothesis. The Law and Administration groups inform an approach between university technology production for business management, allowing us to investigate whether AI has the potential to address a historical demand in the Brazilian development model: the lack of structural couplings between knowledge and the market in Brazil (Miranda, 2011; Mirra, 2009).

Acknowledging the limitations of a single survey, we bring new research directions. Firstly, there is enough data to produce hierarchical and bipartite social networks between research groups and their themes. In this type of social network, there are two independent sets of points, and relations are made from one set to the other (Costa et. al, 2021), in our case, between knowledge areas and their themes. The networks present relations greater than those presented in textual analyses, deepening our analytical character. Secondly, we will be able to find new descriptors that could be incorporated into the new collection in 2024 - after the DGP-CNPq Census of 2023, which we believe encouraged updates of information provided by the groups. The aim is to find the largest possible number of research groups in human and social sciences studying Artificial Intelligence, even capturing topics marginalized by the initial descriptors. With data from different years, we will be able to capture the conformation of this new research ecosystem and track whether indeed artificial intelligence is a new tool within national development and its role in bridging the research and market spheres within the Brazilian innovation ecosystem. The creation of temporal social networks can be a great ally in this process, pointing out the dynamic directions of this scientific field to confirm and create new hypotheses.

Thirdly, the idea of an Artificial Intelligence Observatory is no longer so distant, capable of automatically classifying research groups based on their common interests, bringing the scientific community in AI - especially human and social sciences - closer together. With future collections and the development of the observatory, it is possible to predict how the social networks between the groups will transform. The existence of common terms among research areas, as demonstrated by "organization" in the areas of administration and information science, provides us with directions to relate the areas in social networks, while the temporal dimension will make the similarities and differences in the adoption of these terms clearer. Finally, the national observation provides the starting point to identify specificities of the Brazilian scientific community in human and social sciences in the investigation of AI through comparisons with thematic trends in other countries. Therefore, we will expand the scope of research to observe regional and global trends (sci map) in order to understand the role of Brazilian scientific production in relation to the world and its capacity to promote innovation.

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